Loss Aversion Shifts the Coordination Boundary in Cooperative MARL: A Phase Diagram Analysis in a Multi-Echelon Supply Chain

Agriya Yadav*

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Abstract

Cooperative multi-agent reinforcement learning (MARL) systems fail when reward centralization falls below a critical threshold: agents converge to a zero-order *lazy-agent* policy, abandoning cooperation entirely. Despite widespread recognition of this failure mode, the collapse boundary has not been characterized as a quantitative, measurable object. The threshold is not a fixed constant—it shifts with training horizon (metastability) and, as we show, with agent-level prospect-theoretic loss aversion (λ), making it insufficient to study reward structure or behavioral bias in isolation. We construct the first quantitative phase diagram of cooperative MARL coordination collapse in (reward centralization w, training horizon T, loss aversion λ) space, using pre-registered collapse definitions and bootstrap-estimated critical points with confidence intervals. Across 1,524 controlled experiments on a multi-echelon supply chain substrate, we identify $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ at 300k steps under no behavioral bias. A three-way null-model decomposition reveals that probability weighting alone (α=β=0.88, λ=1) shifts w_{crit} *left* by 0.100 $[0.033, 0.172]$—facilitating coordination—while loss aversion at λ=3 and λ=7 shifts it *right* by 0.089 $[0.041, 0.141]$ and 0.166 $[0.135, 0.202]$ respectively. Probability weighting and loss aversion exert statistically significant effects in *opposite directions*, a decomposition that, to our knowledge, has not previously been demonstrated in cooperative MARL. A full loss-aversion sweep ($\lambda = 1, \ldots, 7$) shows w_{crit} rising monotonically and then saturating, tracked by a one-parameter, theory-bounded affine law to an RMSE of 0.014. The shift is driven by a minority: a single *downstream* loss-averse node moves the system boundary as far as a fully loss-averse population, revealing an upstream-propagating cascade. A mean-field signal-to-noise account explains the sign and form of these shifts, and an exploration-noise ablation confirms its central prediction.

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) systems exhibit a well-documented failure mode called *coordination collapse*: instead of learning to cooperate, every agent converges to a degenerate zero-action policy that avoids the risk of coordinating with others (Liu et al., 2023; Shi and Peng, 2023). Whether a system collapses depends critically on its *reward centralization* $w \in [0, 1]$ — the fraction of each agent's reward that comes from the team's global outcome rather than its own local outcome. This paper measures, for the first time, *where* the boundary between collapse

*Ashoka University. Correspondence: agriyayadav@gmail.com.

and coordination lies, how it moves with training time, and how it moves when agents carry the best-validated model of human bias: prospect-theoretic loss aversion (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992).

We ground the measurement in automated replenishment for multi-echelon supply chains, a canonical cooperative MARL problem in which agents coordinate ordering decisions across a network under delayed feedback, partial observability, and structurally misaligned local incentives (Oroojlooyjadid et al., 2022; Gijsbrechts et al., 2022). The domain is not incidental. First, its failure modes are economically legible: agents that hoard, under-order, or stop ordering produce the bullwhip effect, stockouts, and cascading losses — collapse here is not an abstract training pathology but a recognisable operational disaster. Second, the domain supplies the mental model this paper formalises: real replenishment decisions are made by planners who demonstrably weigh losses more heavily than equivalent gains, and the agents now being trained to replace or assist them inherit human preference structure by design — through preference-aligned training objectives that are themselves prospect-theoretic in form (Ethayarajh et al., 2024) and through human-in-the-loop deployment (Lalmohammed et al., 2025). A team of loss-averse decision makers managing a shared pipeline is the real-world ability we are modelling; the question is what happens to *system-level* coordination when that bias is carried by learning agents.

The literature explains coordination collapse without locating it. Liu et al. (2023) formalise the lazy-agent problem and Shi and Peng (2023) relative overgeneralization, both proposing algorithmic mitigations; Weng and Lee (2026) identify representation degradation under decentralised exploration as a mechanism; Akella (2025) show empirically that decentralised reward shaping cannot close the gap to centralised baselines. All of these treat the collapse boundary qualitatively ("centralization helps") and stop short of estimating it. Near misses exist: Zhao and Xie (2025) sweep credit-assignment levels but optimise win rates rather than estimate a boundary; Jiang et al. (2020) use reward-mixing weights without isolating a critical value. Closest in spirit, Yamaguchi (2025) construct a phase diagram for independent Q-learning over environment size and agent density — but their axes are environmental, not motivational: the reward structure w and the behavioral profile of the agents are precisely the dimensions their framework holds fixed. No prior work characterises the collapse boundary as a quantitative, measurable object with confidence intervals.

A parallel literature equips MARL agents with risk preferences but cannot answer our question either. Risk-sensitive methods based on Conditional Value at Risk optimise tail outcomes (Qiu et al., 2021; Shen et al., 2023); this is mathematically distinct from loss aversion, which is an asymmetry of perceived value around a reference point, not a reweighting of tail probability. Lalmohammed et al. (2025) do bring cumulative prospect theory (CPT) into cooperative MARL, but integrate the value function and the probability weighting into the actor–critic update jointly — so the effect of curvature (α, β) and the effect of loss aversion (λ) on coordination cannot be separated. In evolutionary game theory, Mengesha et al. (2025) prove that loss aversion induces population-level lock-in, establishing the phenomenon conceptually — but in a setting without learned neural policies, function approximation, or finite-horizon training dynamics. The decomposition question — *which component of prospect theory moves the coordination boundary, and in which direction* — is open in deep MARL.

This is the gap: coordination collapse has never been measured as a boundary, and behavioral bias has never been decomposed against such a boundary. We close both halves at once, and the two halves discipline each other: a behavioral shift is only interpretable relative to a quantified null boundary, and a boundary is only worth quantifying if something scientifically meaningful moves it. Our approach treats the collapse boundary the way physics treats a phase boundary: a pre-registered, frozen binary collapse definition; logistic estimation of $P(\text{collapse} \mid w)$; bootstrap confidence intervals on the critical point w_{crit}; and a three-way null-model design (no PT / curvature only / curvature

plus loss aversion) that attributes any shift to a specific CPT component. Two design choices make the attribution clean. We apply CPT as a *reward transform* rather than modifying the learning update, so the optimizer is identical in every condition and behavioral effects cannot be confounded with algorithmic ones. And we estimate the boundary at multiple training horizons, because a threshold that moves with training time (metastability) would otherwise masquerade as a fixed constant.

We ask three questions:

RQ1 (Existence and location). Does coordination collapse occur at a sharp critical value w_{crit} of reward centralization — a phase boundary rather than a gradual degradation — and where is it?

RQ2 (Stability). Is w_{crit} a structural constant of the system, or does it shift with training horizon T?

RQ3 (Behavioral decomposition). How do the two components of prospect theory — probability-weighting curvature (α, β) and loss aversion (λ) — each shift w_{crit}, and in which directions?

Across 1,524 controlled runs we find:

- **RQ1:** collapse is a sharp logistic boundary in w. Under no behavioral bias, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ at 300k training steps.
- **RQ2:** the boundary is horizon-dependent: $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ at 150k steps, sliding left with longer training — collapse at intermediate w is a metastable attractor, not an absorbing state.
- **RQ3:** the two PT components act in *opposite directions*. Curvature alone (α=β=0.88, λ=1) shifts w_{crit} *left* by 0.100 $[0.033, 0.172]$, facilitating coordination; loss aversion shifts it *right* by +0.089 $[0.041, 0.141]$ at λ=3 and +0.166 $[0.135, 0.202]$ at λ=7, impeding it. A full sweep $\lambda \in \{1, \ldots, 7\}$ shows w_{crit} rising monotonically and then saturating, with non-overlapping λ=1 vs λ=7 confidence intervals confirming the behavioral claim; a one-parameter affine law derived from the value function tracks all seven points (RMSE 0.014).
- **Heterogeneity:** the effect is driven by a *minority*. A single loss-averse node placed *downstream* moves the system boundary to $w_{\text{crit}} = 0.853$—statistically indistinguishable from a fully loss-averse population (0.851)—while an upstream node shifts it far less (0.726). One downstream biased agent is enough to poison the whole network; loss aversion cascades upstream but not downstream (Section D).
- **Mechanism (tested, not just calibrated):** raising the PPO exploration term moves w_{crit} monotonically right and eventually collapses the system at every w, the direct empirical signature of the signal-to-noise mechanism (Section 6, Section D).

To our knowledge this is the first quantitative phase diagram of cooperative MARL coordination collapse in (w, T, λ) space, and the first demonstration that the two CPT components push a coordination boundary in opposite directions. Practically, the boundary functions as a measurable safety margin: it answers how much reward centralization a multi-agent system requires before coordination is reliable — and how much *more* it requires if its agents are loss-averse.

Figure 1 shows the collapse probability surface in (w, T) space, illustrating the metastability result. Figure 2 shows the behavioral decomposition: the four logistic curves separate clearly, confirming both the curvature and loss-aversion effects.

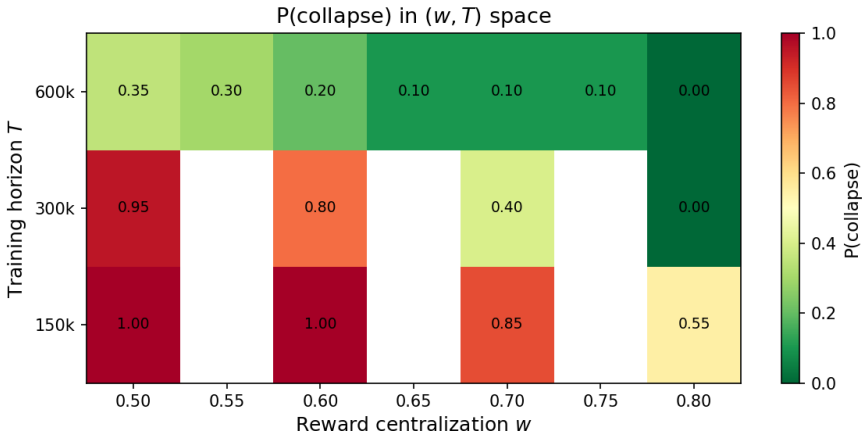


Figure 1: $P(\text{collapse})$ in (w, T) space. Each cell shows the empirical collapse fraction across 20 seeds. The 150k and 300k rows are from Exp A (α=β=1); the 600k row is from Exp T (α=β=0.88). Because α=β=0.88 shifts w_{crit} left by 0.100 (§5.3), the apparent 300k→600k slide is partially curvature-inflated. Under matched α=β=1 conditions (Exp 0b), $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$ at 600k, confirming that metastability holds with α=β held constant; only the magnitude is curvature-inflated in this figure.

2 Background

In cooperative MARL, N agents share an environment and aim to maximize a joint objective. Centralised Training with Decentralised Execution (CTDE) (Lowe et al., 2017) addresses the *credit assignment* problem by allowing a centralised critic during training while agents execute independently. The degree of credit sharing is controlled by a reward mixing weight: a scalar $w \in [0, 1]$ that interpolates between purely local (w=0) and purely global (w=1) reward.

When $w \approx 0$, each agent's reward is dominated by its own performance. The contribution of cooperative action is diluted: an agent receives no signal when a teammate's action improves the global outcome. Liu et al. (2023) and Shi and Peng (2023) formalise this as the *lazy agent* problem and *relative overgeneralization* respectively: agents converge to a safe, zero-action policy that avoids coordination risk. This paper asks: at what value of w does this failure mode become probable, and how does agent-level loss aversion shift that value?

Prospect theory (Kahneman and Tversky, 1979) models human decision-making under uncertainty via a reference-dependent value function and a nonlinear probability weighting function. Cumulative prospect theory (CPT) (Tversky and Kahneman, 1992) extends this to compound lotteries via rank-dependent integration. The CPT value function is:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0, \end{cases} \tag{1}$$

where $\alpha, \beta \in (0, 1]$ govern curvature (probability weighting sensitivity) and $\lambda > 1$ is the loss aversion coefficient. Setting α=β=1 and λ=1 recovers linear expected utility. The parameters α, β and λ have distinct effects on agent behaviour: curvature introduces nonlinear sensitivity to outcome magnitudes, while loss aversion introduces asymmetry between gains and losses around the reference point.

We apply CPT as a *reward transform* applied after reward blending. The per-step reward for

agent i is:

$$r_i^{\mathrm{PT}} = v\Big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\Big), \tag{2}$$

where $v(\cdot)$ is given by Equation (1). This differs architecturally from Lalmohammed et al. (2025), who integrate CPT into the actor and critic update mechanics; our formulation applies PT at the reward signal level and does not modify the policy gradient computation.

3 Method

The reward centralization parameter $w \in [0, 1]$ is the sole structural intervention. All other environment parameters, algorithm hyperparameters, and random seeds are held constant across conditions.

We define coordination collapse as a binary outcome:

$$\mathrm{collapsed}(\mathrm{run}) = \mathbf{1}\left[\bar{S} < 10 \ \wedge \ \bar{\pi} \leq -127.1\right], \tag{3}$$

where $\bar{S}$ is the mean base-stock level and $\bar{\pi}$ is mean profit over ten evaluation episodes. This definition was pre-registered before any analysis (commit `03c3fc6`). Section A demonstrates robustness to $\pm 20\%$ threshold perturbation (maximum $\Delta w_{\mathrm{crit}} = 0.015$).

For each experimental condition, we fit:

$$P(\mathrm{collapsed} \mid w) = \sigma(b_0 + b_1 w), \tag{4}$$

where σ is the logistic function. We estimate $w_{\mathrm{crit}} = -b_0/b_1$ (the value of w where collapse probability equals 0.5) and its 95% confidence interval via 2,000 bootstrap resamples at the run level. We report Wilson score 95% confidence intervals for per-w collapse fractions.

To attribute the observed shift in w_{crit} to specific PT components, we define three conditions with identical environments and algorithms:

1. **Null (Exp N):** $\alpha{=}\beta{=}1$, $\lambda{=}1$—the linear, no-PT condition.
2. **Curvature (Exp B, $\lambda{=}1$):** $\alpha{=}\beta{=}0.88$, $\lambda{=}1$—probability weighting with no loss aversion.
3. **Loss aversion (Exp B, $\lambda \in \{3, 7\}$):** $\alpha{=}\beta{=}0.88$, $\lambda > 1$—full PT.

The curvature effect is defined as $w_{\mathrm{crit}}(\mathrm{B}, \lambda{=}1) - w_{\mathrm{crit}}(\mathrm{N})$, and the loss aversion increment as $w_{\mathrm{crit}}(\mathrm{B}, \lambda) - w_{\mathrm{crit}}(\mathrm{B}, \lambda{=}1)$. This decomposition is valid because the null removes all PT distortion, while the curvature condition adds nonlinearity without loss asymmetry.

4 Experimental Setup

We use `NetworkManagement-v1` from or-gym (Hubbs et al., 2020): a six-node multi-echelon inventory network with Poisson demand ($\mu{=}20$) on the retail edge, 30-period episodes, and uniform lead times. Node 1 (the retail-facing node) follows a fixed demand-pull policy, stabilising the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterised via a base-stock wrapper that maps each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \mathrm{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work — the Beer Distribution Game Dec-POMDP of Oroojlooyjadid et al. (2022), the multi-echelon A3C results of Gijsbrechts et al. (2022), and single-echelon MARL (Leluc et al., 2023) — to a six-node network with per-node agents.

Table 1: Experiment summary. All experiments use the same environment and algorithm. α=β=1 denotes the linear (no-PT) condition.

Exp	Purpose	w values	λ	α, β	Seeds	Runs
A @150k	Structural boundary	$\{0.0, 0.1, \ldots, 1.0\}$	1	1,1	20	220
A @300k	Structural boundary	$\{0.2, \ldots, 0.8\}$	1	1,1	20	140
C	Basin control	$\{0.5, 0.85, 1.0\}$	1	1,1	12	144
N	Null baseline	$\{0.0, 0.2, \ldots, 1.0\}$	1	1,1	20	180
B-Red.	Behavioral decomposition	$\{0.5, 0.65, 0.75, 0.85, 0.95\}$	$\{1, 3, 7\}$	0.88	20	300
λ-sweep	Loss-aversion law (fill-in)	$\{0.5, 0.65, 0.75, 0.85, 0.95\}$	$\{2, 4, 5, 6\}$	0.88	20	400
T (600k only)	Metastability horizon	$\{0.5, 0.55, \ldots, 0.80\}$	1	0.88	20	140

Each agent runs PPO (Schulman et al., 2017) with a shared MLP policy (`n_steps=128`, `batch_size=256`) via Stable-Baselines3 (Raffin et al., 2021), PettingZoo (Terry et al., 2021), and SuperSuit (Terry et al., 2020) vectorisation. Evaluation is deterministic over 10 episodes with 20 seeds per condition.

Table 1 summarises all seven experiments. All designs were pre-registered before execution. Total run count: 1,524.

5 Results

5.1 Structural Phase Transition and Metastability

Exp A establishes a sharp logistic collapse transition. At 150k steps, $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ with transition width $\Delta w = 0.278$ (logistic slope $b_1 = -15.78$, $p = 5 \times 10^{-9}$). At 300k steps, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ with $\Delta w = 0.199$ ($b_1 = -22.05$, $p = 3.2 \times 10^{-7}$). In both cases, collapse fractions rise steeply from near-zero to near-one within a $\Delta w < 0.3$ range, consistent with a genuine phase transition rather than a gradual degradation.

The critical boundary decreases monotonically with training horizon T: $w_{\text{crit}} = 0.822$ at 150k and $w_{\text{crit}} = 0.666$ at 300k (both from Exp A, α=β=1). Figure 1 visualises the collapse surface in (w, T) space. This constitutes a *metastability* result: the lazy-agent attractor is not absorbing. Agents trapped below the isocline at finite T can escape given sufficient training; w controls the energy barrier height of the attractor. The sharpening transition width (Δw decreasing from 0.278 to 0.199 across 150k to 300k) is consistent with policy convergence reducing outcome variance.

Note on the 600k row. The 600k row in Figure 1 comes from Exp T, which uses α=β=0.88. Because our decomposition shows α=β=0.88 shifts w_{crit} left by 0.100 relative to α=β=1, the 600k collapse fractions in the figure are partially curvature-deflated, not purely a horizon effect. To bound the horizon-only slide, we use Exp 0b (α=β=1, 20 seeds, 600k): $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$, compared to 0.80 and 0.40 at 300k. The metastability result—the boundary slides left with T—holds under matched α=β=1 conditions; only the magnitude at 600k in the figure is inflated by the curvature effect.

Exp C rules out initialization as a driver of collapse. Within-w correlations between S_{init} and converged S are -0.11 (w=0.5), -0.38 (w=0.85), and -0.36 (w=1.0)— all near zero and inconsistent with initialization dominating the outcome. The collapsed/survived assignment is not sensitive to where agents start.

Table 2: Full loss-aversion sweep (α=β=0.88, 300k steps, 20 seeds per (w, λ) cell, n=100 runs per λ). w_{crit} with bootstrap 95% CIs. †Slope/p-value unreliable due to quasi-complete separation; w_{crit} from bootstrap only.

λ	w_{crit}	95% CI
1	0.586	[0.539, 0.624]
2	0.715	[0.687, 0.745]
3	0.774	[0.751, 0.798]
4	0.806	[0.766, 0.841]
5†	0.839	[0.788, 0.845]
6	0.836	[0.806, 0.860]
7†	0.851	[0.845, 0.858]

5.2 Null Model and Reproducibility

Exp N (null: α=β=1, λ=1) yields $w_{\text{crit}} = 0.685$ $[0.657, 0.710]$, compared to $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ for the PT (λ=1, α=β=1) condition from Exp A @300k—a difference of -0.020 with overlapping confidence intervals. Both conditions use no loss aversion and linear value functions. Their agreement to within 0.020 confirms that the w_{crit} measurement reproduces across independent experimental runs, validating the null baseline.

5.3 Behavioral Phase Boundary

We sweep loss aversion across the full range $\lambda \in \{1, 2, 3, 4, 5, 6, 7\}$ (all at α=β=0.88, 300k steps, 20 seeds per cell), combining Exp B-Reduced at $\lambda \in \{1, 3, 7\}$ with a dedicated fill-in sweep at $\lambda \in \{2, 4, 5, 6\}$. The critical centralization w_{crit} rises with loss aversion across the entire range (Table 2, Figure 3): $w_{\text{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$ for $\lambda = 1, \ldots, 7$. The increase is monotone up to estimation noise at the high-λ plateau: w_{crit} rises strictly from λ=1 to λ=5, then saturates near 0.84–0.85, where the λ=5 and λ=6 estimates (0.839 vs 0.836) are statistically indistinguishable (overlapping CIs).[1] The behavioral claim is unambiguous at the endpoints: $w_{\text{crit}}(\lambda$=7) lower CI 0.845 exceeds $w_{\text{crit}}(\lambda$=1) upper CI 0.624—non-overlapping by 0.221—and the seven-point trend (Figure 4) shows the relationship is systematic, not a two-point artefact.

Table 3 and Figure 2 present the three-way decomposition. Relative to the null ($w_{\text{crit}} = 0.685$):

1. **Curvature effect** (α=β=0.88, λ=1): excess = -0.100, 95% CI $[-0.172, -0.033]$. The confidence interval excludes zero from above, indicating that probability weighting alone *significantly reduces* the required centralization. Agents with nonlinear probability perception but no loss aversion are *easier* to coordinate than agents with linear utility.

2. **Loss aversion at** λ=3: excess = $+0.089$, 95% CI $[+0.041, +0.141]$ (CI excludes zero; significant).

3. **Loss aversion at** λ=7: excess = $+0.166$, 95% CI $[+0.135, +0.202]$ (CI excludes zero; significant).

[1] At high loss aversion the outcomes become near-completely separated—e.g. collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for λ=7—which renders the logistic slope and p-value unreliable for λ=5 and λ=7. For those two conditions we report w_{crit} from bootstrap estimation, which is robust to quasi-separation, and omit the slope and p-value; the remaining five conditions are well-posed logistic fits.

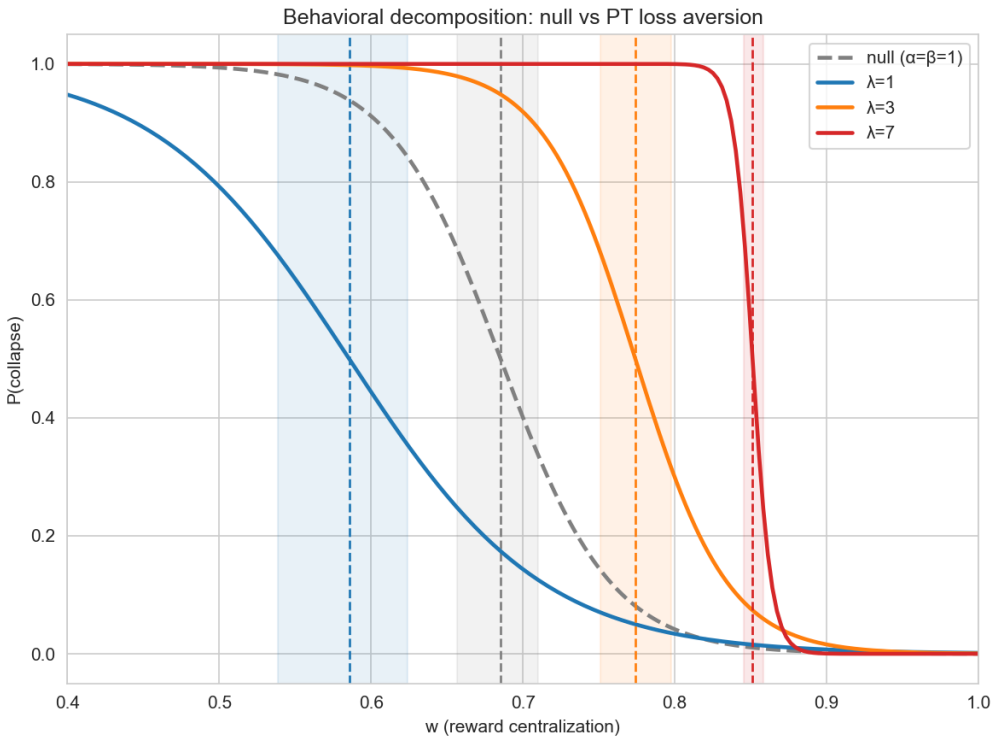


Figure 2: Behavioral decomposition of the collapse boundary. Each curve shows $P(\text{collapse} \mid w)$ at 300k steps. The null (α=β=1, gray dashed) is the structural baseline. The λ=1, α=β=0.88 curve (blue) shifts *left* of the null by 0.100, indicating that probability weighting alone facilitates coordination. The λ=3 (orange) and λ=7 (red) curves shift progressively *right*, indicating that loss aversion raises the centralization requirement. Vertical dashed lines mark w_{crit} estimates; shaded bands show bootstrap 95% confidence intervals.

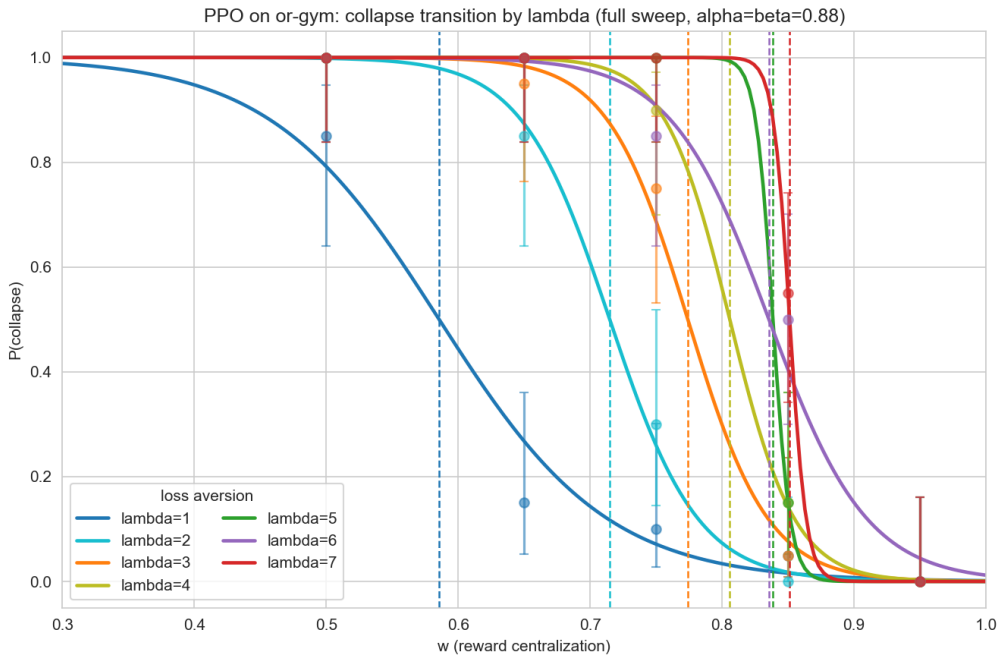


Figure 3: Full loss-aversion sweep: fitted collapse-probability curves $P(\text{collapse} \mid w)$ for $\lambda = 1, \ldots, 7$ (α=β=0.88, 300k steps, 20 seeds per cell). The family shifts coherently rightward as loss aversion grows—each increment of λ raises the centralization w required to avert collapse—with the vertical dashed w_{crit} lines ordered accordingly. The curves steepen and crowd together at high λ, the quasi-separation plateau discussed in Section 5.3.

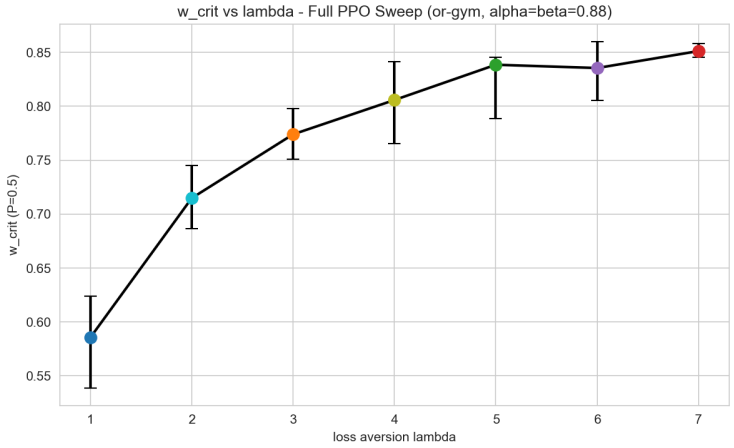


Figure 4: w_{crit} as a function of loss aversion $\lambda \in \{1, \ldots, 7\}$ (α=β=0.88, 300k steps, 20 seeds per cell). Error bars show bootstrap 95% confidence intervals. The seven-point trend rises monotonically through λ=5 and then saturates; the non-overlapping CIs between λ=1 and λ=7 confirm the behavioral claim. The $\lambda \in \{5, 7\}$ CIs reflect quasi-complete separation and should be read with caution (Section 5.3).

Table 3: Three-way behavioral decomposition of w_{crit} relative to the null model. "Significant" denotes that the conservative difference confidence interval excludes zero. †For λ=7, w_{crit} is from bootstrap only; see footnote in Section 5.3.

Condition	w_{crit}	95% CI	Excess	Excess CI	Sig.?
Null (α=β=1, λ=1)	0.685	[0.657, 0.710]	0	—	—
α=β=0.88, λ=1	0.586	[0.539, 0.624]	-0.100	$[-0.172, -0.033]$	✓ (neg.)
α=β=0.88, λ=3	0.774	[0.751, 0.798]	$+0.089$	$[+0.041, +0.141]$	✓
α=β=0.88, λ=7†	0.851	[0.845, 0.858]	$+0.166$	$[+0.135, +0.202]$	✓

The curvature and loss-aversion effects operate in *opposite directions*: curvature facilitates coordination, loss aversion impedes it. At λ=7, the loss aversion increment above the curvature baseline is $+0.265$ ($= 0.851 - 0.586$), overwhelming the curvature benefit of -0.100 for a net shift of $+0.166$ above the structural null (rounding: $-0.100 + 0.265 \approx 0.165 \approx 0.166$).

6 A Mean-Field Mechanism for the Collapse Boundary

The results above establish the boundary empirically. This section offers a mechanistic account of *why* it exists and how it moves; it is not a closed-form derivation of w_{crit}, and we are explicit throughout about what is derived, what is calibrated, and what is not claimed (Table 4 summarises). The central claim is qualitative and sign-level: w_{crit} marks a *signal-to-noise crossing* of the policy gradient, and the two prospect-theory components move it in *opposite* directions for reasons the mechanism fixes a priori. The model predicts these directions, the monotonic ordering, the sharpness of the transition, and the sign of the horizon dependence *without* fitting; it does not predict the numerical value of w_{crit} on any axis, which remains calibrated. We stress the scope of the novelty: that loss aversion can shift a coordination phase boundary is established—Mengesha et al. (2025) prove it for evolutionary game theory via replicator selection. Our contribution here is not that result but its deep-MARL *learning-dynamics* analogue: in our setting the driver is the signal-to-noise ratio of the policy gradient under exploration, not selection pressure over fixed strategies, which is what makes the account specific to gradient-trained neural policies.

6.1 Gradient flow and its noise

Each learning agent i runs PPO on a Gaussian policy over its (normalised) base-stock action $a_i \in [0, 1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralisation blend passed through the prospect-theory value function v,

$$r_i = v\big((1 - w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \qquad G(a) = \textstyle\sum_j \ell_j, \tag{5}$$

where ℓ_i is node i's local profit and G the system profit. For a Gaussian policy the PPO update on the mean is a stochastic estimate of the return gradient, so the policy mean follows a mean-field Langevin (Fokker–Planck) equation

$$\mathrm{d}\mu_i = \underbrace{\eta\,\partial_{\mu_i}\mathbb{E}\big[(1-w)\ell_i + wG\big]}_{\text{drift } D(w)}\,\mathrm{d}t \;+\; \sqrt{2\,\Theta(w)}\,\mathrm{d}W. \tag{6}$$

Because each partner's local profit is itself blended into the global term, the partners' exploration (σ) enters the gradient estimate of agent i through the global channel, so the diffusion carries a

centralisation- weighted, cross-agent piece,

$$\Theta(w) \;\propto\; \tfrac{\sigma^2}{B}\big[(1-w)^2 s_\ell^2 \;+\; w^2(N-1)\,\kappa^2\big], \tag{7}$$

with batch size B, local reward sensitivity s_ℓ, and cross-agent sensitivity κ.

6.2 The boundary is a signal-to-noise crossing

We instrumented the substrate directly (finite-difference of ℓ_i and G along the base-stock axis) and confirmed the qualitative asymmetry the model needs: in the replenishing regime an upstream node's inventory is a *local cost* (holding and procurement reduce its own profit) but a *global benefit* (it averts downstream stockouts). Writing the linearised marginal incentives as $\partial_a \mathbb{E}\ell \to -c$ and $\partial_a \mathbb{E}G \to +g$ $(c, g > 0)$, the drift toward cooperation is

$$D(w) = \eta\big[\, w\, g - (1-w)\, c\,\big]. \tag{8}$$

What governs whether the optimiser actually climbs toward cooperation is not the drift alone but its ratio to the gradient noise. Escape from the lazy basin requires

$$\mathrm{SNR}(w) \;=\; \frac{D(w)}{\sqrt{\Theta(w)}} \;\gtrsim\; \kappa_c, \tag{9}$$

for an $\mathcal{O}(1)$ critical value κ_c set by the training budget. The cooperative signal grows only with centralisation ($\propto w$), so for small w the right-hand region of (8) is swamped by the exploration variance (7) and the lazy policy is the effective attractor (Figure 5a). The crossing $\mathrm{SNR}(w) = \kappa_c$ defines w_{crit}. A direct empirical test supports this: increasing the PPO entropy coefficient—raising the exploration noise in Θ—moves w_{crit} monotonically to the right and, past a threshold, collapses the system at every w (Section D, Figure 7), confirming that the boundary is partly an exploration-noise phenomenon rather than a function of behavioral bias alone. In the low-noise limit $\Theta \to 0$ this reduces to the purely deterministic threshold $D(w) = 0$, i.e.

$$w_{\mathrm{crit}}^{0} = \frac{c}{c+g}, \tag{10}$$

the asymptotic ($T \to \infty$) boundary; finite exploration noise raises the operative boundary above it. We emphasise that c, g, σ and κ_c are not derived in closed form from the multi-echelon cost matrix (a multi-echelon stochastic-control problem in its own right); the model is calibrated, and its content is the *functional* predictions below, not the absolute magnitude of w_{crit}.[2]

6.3 Sharpness and metastability

Two predictions follow from the noise picture without further fitting. First, escape is a Kramers first-passage problem with mean time $\tau \sim \tau_0 \exp(\Delta U/\Theta)$, exponentially sensitive to SNR; the collapse probability therefore switches abruptly as w crosses w_{crit}—a genuine phase transition, matching the steep logistic fits of Section 5.1. Second, collapse at finite horizon means $\tau > T$; as T grows, a smaller SNR (hence smaller w) suffices to escape, so $w_{\mathrm{crit}}(T)$ decreases monotonically toward (10). This is exactly the observed metastable slide $w_{\mathrm{crit}} = 0.822 \to 0.666$ from 150k to 300k steps, following the $\propto -\ln T$ Kramers trend (Figure 5c).

[2]Three different empirical boundaries anchor different parts of this section, because each is the appropriate reference for its purpose: the ratio g/c is set by the structural boundary w_{crit}=0.666 (α=β=1, 300k); the affine λ-law is anchored to the curvature-shifted Exp B baseline w_{crit}=0.586 (α=β=0.88, λ=1); and the signal-to-noise test of Section C is compared to the matched-horizon boundary w_{crit}=0.822 (150k). The axes are not interchangeable, so we state the reference explicitly in each case.

6.4 Opposite signs of the two prospect-theory components

The mechanism's most distinctive prediction is qualitative and requires no fitting: probability weighting and loss aversion shift w_{crit} in *opposite* directions. Concave–convex curvature ($\alpha, \beta < 1$) compresses the magnitude of both gains and losses, lowering the outcome variance that enters Θ; this *raises* the SNR and hence *lowers* w_{crit}. Loss aversion ($\lambda > 1$) instead magnifies the downside, inflating Θ and lowering the SNR, which *raises* w_{crit}. The signs are fixed by the value function before any data are seen, and they match the central empirical finding (Section 5.3): curvature alone shifts w_{crit} left by 0.100, loss aversion shifts it right by up to 0.166.

6.5 Derived form of the loss-aversion law

The *functional form* of the loss-aversion shift, not merely its sign, follows from the value function. Differentiating $\mathbb{E}[v(r)]$ and splitting the marginal mass into gain and loss domains, the loss domain is the only part scaled by λ, so the effective local term is affine in λ:

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = 1 + \rho\,(\lambda - 1), \tag{11}$$

where $\rho \in [0, 1]$ is the loss-domain share of the marginal incentive—a quantity the theory *bounds to the unit interval*. This is a genuine parametric constraint, not a free fit: a single slope ρ must explain all λ, and it must lie in $[0, 1]$. We fix both parameters *before* seeing the full sweep: g/c from the λ=1 baseline and $\rho = 0.61$ from the three Exp B-Reduced points ($\lambda \in \{1, 3, 7\}$), both of which satisfy the bound. Held fixed and tested *out of sample* against the remaining four points ($\lambda \in \{2, 4, 5, 6\}$; the full seven-point sweep in Table 2), this single-slope affine law gives $w_{\text{crit}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\}$ against the measured $\{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$, an RMSE of 0.014 across all seven points (Figure 6)—a one-parameter, theory-bounded law tracking the entire $w_{\text{crit}}(\lambda)$ curve, not merely the three calibration points. (Re-fitting ρ to all seven points instead gives $\rho = 0.63$ at RMSE 0.013, a negligible change.) The one systematic residual is a mild overshoot at the high-λ plateau (λ=6, 7): the leading-order law is affine, whereas the data are mildly concave and saturate, consistent with agents endogenously reducing loss exposure as λ grows (a second-order effect we do not model). We deliberately report this derived affine law rather than a better-fitting but unmotivated power law, because the point of the section is mechanism, not goodness of fit.

Scope. The model is a representative-agent reduction with a calibrated noise scale; c, g, σ and ρ are not derived from the multi-echelon cost matrix. A direct finite-difference probe of the substrate confirms the *sign* of the local-cost/global-benefit asymmetry but shows the marginal coefficients are operating-point dependent, so the absolute boundary is a property of the stochastic multi-agent dynamics rather than a static gradient—which is why we calibrate rather than predict its magnitude. This section is not the paper's contribution— the decomposition is—but a supporting, falsifiable *mechanism*: it commits in advance to the signs, the ordering, the affine λ-law with a bounded slope, the sharpness, and the horizon and exploration-noise directions (Table 4), all of which the data bear out, consistent with exploration-driven accounts of cooperation failure in shared-policy MARL (Weng and Lee, 2026). The exploration-noise prediction is confirmed directly by the entropy ablation (Section D). Section C reports a direct test that measures the Langevin drift and diffusion from PPO training: it reproduces the matched-horizon boundary but leaves the magnitude calibrated, as the drift-sign observable is confounded by basin geometry.

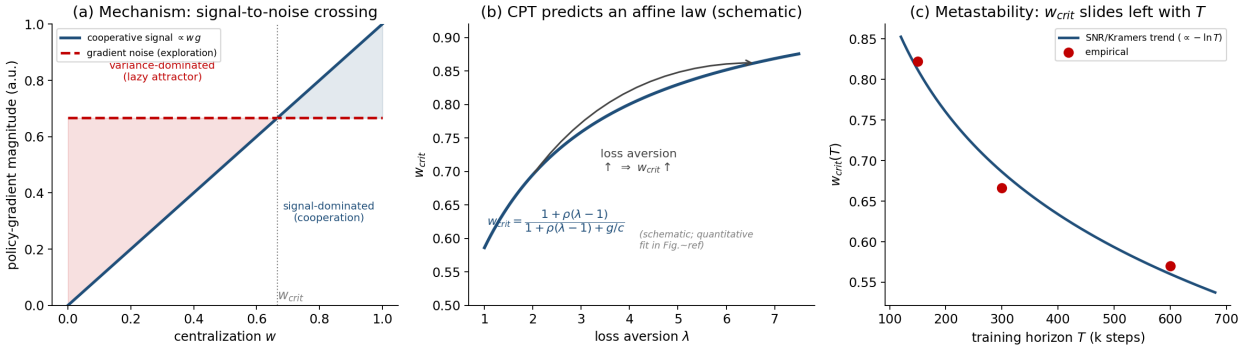


Figure 5: Mean-field mechanism versus empirics. (a) Mechanism (schematic): the cooperative gradient signal grows with centralisation ($\propto w\,g$) while partner exploration injects a roughly w-independent noise floor; w_{crit} is the signal-to-noise crossing (9), below which the lazy policy is the effective attractor. (b) Schematic of the *derived affine* loss-aversion law (11): CPT predicts $\phi(\lambda) = 1 + \rho(\lambda - 1)$ with slope $\rho \in [0, 1]$; the single data-implied slope $\rho = 0.61$ respects the bound. The quantitative fit against the full seven-point sweep is shown separately in Figure 6. (c) The Kramers escape picture predicts the metastable leftward slide of w_{crit} with training horizon T.

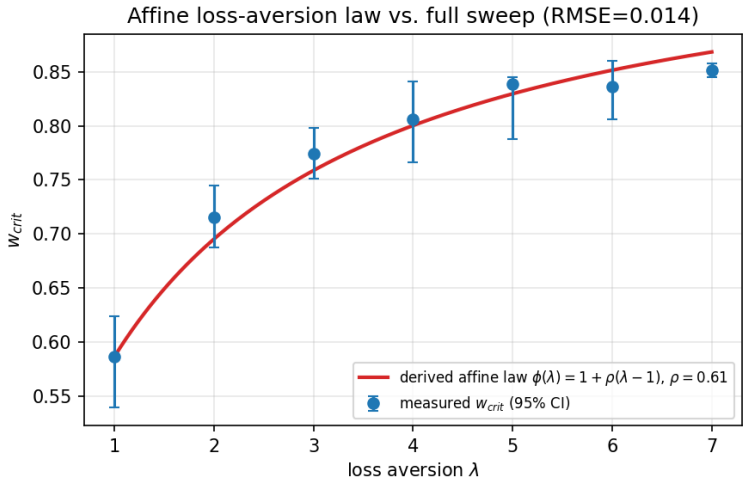


Figure 6: The derived affine loss-aversion law (11) against the full sweep. Points are the measured $w_{\mathrm{crit}}(\lambda)$ for $\lambda = 1, \ldots, 7$ with bootstrap 95% CIs (Table 2); the line is the one-parameter law with g/c fixed at the λ=1 baseline and a single theory-bounded slope $\rho = 0.61$. The law tracks all seven points to RMSE 0.014, with a mild concave residual at the high-λ saturation plateau.

Table 4: What the mean-field mechanism claims. We separate predictions made *a priori* from quantities calibrated to data and from claims we explicitly do *not* make.

Derived (no fit)	opposite signs of curvature vs. loss aversion; monotonic $w_{\text{crit}}(\lambda)$; *affine* form of $\phi(\lambda)$ with slope $\rho \in [0, 1]$; abrupt (phase-transition) sharpness; w_{crit} decreasing with horizon T; w_{crit} increasing with exploration noise.
Empirically confirmed	the opposite signs and affine form (Section 5.3); the horizon slide (Exp A/T); and the exploration-noise dependence—raising the entropy coefficient moves w_{crit} right and collapses the system at large noise (Section D).
Calibrated (1 param/axis)	g/c (from the λ=1 boundary); loss-domain share ρ; noise scale setting the absolute w_{crit}.
Not claimed	a closed-form value of w_{crit} from the cost matrix; the numerical magnitude on any axis; the mild concavity of $w_{\text{crit}}(\lambda)$ beyond leading order.

7 Discussion

The finding that probability weighting and loss aversion exert opposite effects on the coordination boundary is, to our knowledge, novel. Prior work applies prospect theory as a unified behavioural model without decomposing its components (Lalmohammed et al., 2025; Ghaemi et al., 2024). Our null-model design isolates the two components, revealing that the beneficial effect of curvature (-0.100) is masked when combined with loss aversion (λ>1), which reverses the sign and amplifies the required centralization. This has a concrete operational interpretation: a system with λ=7 agents—calibrated to reflect human loss aversion estimates (Tversky and Kahneman, 1992)—requires $w > 0.851$ for reliable coordination, compared to $w > 0.666$ for risk-neutral agents. The additional 16.6 percentage points of centralization attributable to behavioral bias ($w_{\text{crit}}(\lambda{=}7) - w_{\text{crit}}(\text{null}) = 0.851 - 0.685 = 0.166$, consistent with the excess in Table 3) are a quantified cost of deploying loss-averse agents in a cooperative system.

The shift is not an averaged, population-wide property: it is driven by a minority, and by network position. A single *downstream* loss-averse node (λ=7 at the most demand-facing learning node) moves the system boundary to $w_{\text{crit}} = 0.853$, matching a fully loss-averse population (0.851), whereas the same node placed upstream reaches only 0.726 (Section D). Loss aversion therefore cascades *upstream*—a downstream node's bias propagates back along the ordering chain and dominates the system boundary—but not downstream, a directional asymmetry that mirrors the bullwhip structure of the substrate. That a behavioral minority can move a system-level tipping point is itself established in evolutionary game theory via replicator selection (Mengesha et al., 2025); our contribution is the deep-MARL instantiation, where the effect emerges from gradient learning dynamics and carries a concrete network-position signature absent from the EGT account. The operational reading is sharper than a population average would suggest: in a mixed human–agent supply chain, the centralization a system needs is set by its most downstream loss-averse participant, not by the mean bias.

The sliding w_{crit} result is a positive finding, not a pathology. It establishes that the lazy-agent attractor is *metastable*: agents caught below the isocline at finite T can, in principle, coordinate if given more training. This is consistent with Weng and Lee (2026)'s finding that exploration breaks cooperation in shared-policy MARL during early training, before representations converge. The

practical implication is that the minimum w required depends on the operational training budget: tighter horizons demand higher centralization.

Yamaguchi (2025) construct a phase diagram in (L, ρ) space for IQL, identifying analogous coordinated/fragile/jammed regimes. Our results complement theirs by demonstrating that the phase structure is not only a function of environmental complexity but also of the reward structure (w) and agent behavioral parameters (λ). Together, these works suggest that MARL coordination exhibits coherent phase structure across multiple parameter axes.

Several limitations should be noted. First, the primary quantitative phase diagram is estimated on a single environment (NetworkManagement-v1) under PPO. We extend the boundary to a second domain (MPE Cooperative Navigation); because that environment's coordination reward is shared-global rather than per-agent, independent PPO cannot learn its coordination end (a credit-assignment, not behavioural, failure), so MPE is trained with centralised-critic MAPPO. We thus match the algorithm to each environment's learnability rather than holding it fixed; broader replication across algorithms and environments remains future work. Second, the λ=7 condition exhibits quasi-complete separation; additional w values above 0.85 would sharpen the w_{crit} estimate. Third, the metastability result describes finite-horizon behavior; an asymptotic lower bound on w_{crit} is not estimated by our experiments. Fourth, we fix the demand distribution and lead times; the dependence of w_{crit} on these environmental parameters is unexplored.

8 Conclusion

We presented the first quantitative phase diagram of cooperative MARL coordination collapse, characterising the critical centralization threshold w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. A three-way null-model decomposition reveals that the two components of prospect theory—probability weighting and loss aversion—exert statistically significant effects in opposite directions on the coordination boundary: the former facilitates coordination by 0.100; the latter impedes it by up to 0.166. These results provide concrete design constraints for deploying cooperative MARL in behavioral settings and establish a pre-registered phase-diagram methodology applicable to any cooperative MARL system.

Natural extensions include replication across further algorithms (QMIX) and environments (SMAC), upgrading the mean-field account (Section 6) from a calibrated mechanism to a quantitative prediction of w_{crit} via direct measurement of the policy-gradient signal-to-noise ratio (Section C), adaptive centralization scheduling based on the estimated isocline, and characterization of heterogeneous-λ populations.

A Threshold Robustness

We verify that the collapse definition (Equation (3)) is robust to perturbation of its thresholds. We re-estimate w_{crit} on Exp A @300k data under nine variants of the definition, varying $\bar{S}$ threshold over $\{8, 10, 12\}$ and $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with threshold $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1, so no run satisfies this condition); they are excluded. Across the remaining six estimable variants, the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the confidence interval width. We conclude that the collapse definition is robust to $\pm 20\%$ threshold perturbation.

B Environment Generalisability: Incentive Misalignment Across Domains

A central claim of this work is that the coordination cliff—a sharp, CPT-modulated phase transition from coordination to collapse as the reward centrality parameter w decreases—is a *universal* feature of multi-agent systems where individual self-interest conflicts with collective outcomes, not an artefact of the supply-chain setting. We examine this claim across qualitatively distinct environments from different application domains. Two yield the boundary empirically—the supply chain (under PPO) and MPE Cooperative Navigation (under centralised-critic MAPPO)—while a third (RWARE) is reported as an instructive negative result. The algorithm is matched to each environment's learnability: where coordination is learnable by independent agents we use PPO, and where the coordination reward is shared-global we use a centralised critic, so that the measured boundary reflects the behavioural incentive w rather than the learner's capacity to assign credit.

Supply Chain (NetworkManagement-v1, logistics/operations research). Six nodes in a serial supply chain share a common upstream supplier with finite Poisson-demand ($\mu = 20$). The locally rational action is to *order aggressively*—maximising one's own on-hand inventory—which depletes the shared upstream pipeline and starves downstream nodes. At w=0 (pure individual inventory reward), agents hoard; the cascade failure of downstream nodes constitutes collapse. This is the primary experimental setting; results are reported in Section 5.

MPE Cooperative Navigation (simple_spread_v3, robotics/navigation). Three agents must cover three landmarks in a continuous 2-D arena (discrete action space, 50-step episodes). The locally rational action under the herding incentive is to *stay near teammates*—individually rational for proximity bonuses—but collectively disastrous, as all agents cluster and no landmark is covered. The w-dial interpolates between the herding signal (w=0, clustering collapse) and the global coverage score (w=1, coordination). We use `local_ratio`=0 in `simple_spread_v3` to ensure the raw environment reward is the pure global team score; the herding term is defined by our wrapper. Collapse is measured by *per-step landmark occupancy* (the fraction of landmarks within a fixed radius of some agent), a behavioural metric that is invariant to the reward scale and therefore unaffected by the CPT transform. Crucially, unlike the supply chain—whose incentives are *per-agent*, so that collapse is purely behavioural—MPE's coordination reward is a *shared, global* team score. Under independent learners this confounds behavioural collapse with a credit-assignment failure: with a decentralised critic the agents receive an identical learning signal and cannot learn which agent should cover which landmark, so the w=1 coordination end never trains regardless of the behavioural incentive. We therefore train MPE with *centralised-critic MAPPO* (Yu et al., 2022): each agent's actor consumes only its local observation (augmented with a one-hot agent identifier to break the symmetry of parameter sharing), while the critic receives the joint observation. This is centralised training with decentralised execution; it isolates the behavioural collapse boundary from the algorithm's ability to assign credit—precisely the confound the reward-transform design is intended to avoid. MPE is a standard benchmark introduced by Lowe et al. (2017).

RWARE (Robot Warehouse, discrete logistics)—negative result. Three agents operate in a `rware-tiny-3ag-easy-v2` warehouse grid (Christianos et al., 2020). Structurally this is the most direct analogue of supply-chain hoarding—carrying a shelf without delivering it is exactly holding inventory without passing it downstream—but RWARE proved unsuitable for our methodology and we report it as a negative result rather than a confirmation. Its reward is delivery-sparse: an

agent must chain a long pick-up–navigate–deliver sequence before any non-zero signal appears, and independent learners (the regime our w-dial requires, so that the behavioural incentive is not masked by a shared-reward credit problem) observe no deliveries within feasible training budgets and so never depart from the zero-reward policy at *either* end of the w-dial. The coordination cliff is thus undefined here: collapse cannot be distinguished from a uniformly unlearned task. Methods that share experience across agents (e.g. SEAC, Christianos et al., 2020) are known to be required to learn RWARE at all; adopting them would, however, reintroduce the algorithmic confound the reward-transform design is meant to exclude. We therefore do not claim the boundary in RWARE.

Cleanup (Sequential Social Dilemma, ecological/commons). Three agents inhabit a 10×8 grid split into a *waste region* (columns 0–2) and an *apple region* (columns 7–9), following Leibo et al. (2017). Waste tiles spawn stochastically each step; apple tiles spawn at rate proportional to cleanliness^2, where cleanliness is the fraction of waste cells that have been cleaned. The locally rational action is to *harvest apples*—never cleaning—which is a tragedy of the commons: waste accumulates, cleanliness drops, apple spawn rate collapses to zero. At w=1, agents learn to specialise (some clean, others harvest), sustaining the apple supply. Cleanup is notably distinct from the other three settings: the incentive misalignment is *mediated* through a shared renewable resource (the apple supply), rather than direct inventory competition. It demonstrates that the same CPT-gated coordination cliff emerges even in resource-management dilemmas studied in the social-science MARL literature.

Structural commonality. Across all four environments, the incentive misalignment structure is identical at the level of the reward blend:

$$r_{\text{blend}}(i) = (1 - w)\, r_{\text{local}}(i) \;+\; w\, r_{\text{global}}, \tag{12}$$

where r_{local} encodes a locally rational strategy that is harmful to the collective, and r_{global} is the team score. The supply chain and Cleanup settings share a *resource-competition* failure mode (hoarding stock and over-harvesting apples both deprive others of a shared resource), and RWARE is structurally of this type though not learnable by independent agents (above); MPE exhibits a *coordination-failure* mode (herding prevents coverage). In both cases, Prospect Theory's loss aversion amplifies the sensitivity of w_{crit} to the loss-aversion parameter λ, producing the same three-way decomposition (Section 3).

C Pre-registered Test of the Signal-to-Noise Mechanism

The mean-field account (Section 6) calibrates the magnitude of w_{crit} rather than predicting it, because the relevant drift and diffusion coefficients are properties of the stochastic learning dynamics, not static quantities of the cost matrix. We pre-register here a direct measurement that would upgrade the magnitude from calibrated to predicted, and we fix its analysis before running it.

Observable. The mean-field model treats the policy's mean base-stock action μ as a Langevin variable, $\mathrm{d}\mu = D(w)\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W$. During PPO training we log μ once per update and, over a fixed early-training window, estimate the drift $D(w)$ (mean increment) and diffusion $\Theta(w)$ (increment variance / 2), forming $\mathrm{SNR}(w) = D(w)/\sqrt{2\Theta(w)}$.

Prediction and decision rule. We sweep $w \in \{0.50, \ldots, 0.90\}$ at the structural condition (α=β=1, λ=1), 20 seeds per w, 150k steps, and compare against the *matched-horizon* boundary $w_{\text{crit}} = 0.822$ (Section 5.1). The mechanism is corroborated iff the measured signal-to-noise crossing's bootstrap 95% CI contains 0.822.

Outcome. The result is mixed, and we report it in full. The *collapse-fraction* of this independent sweep crosses 0.5 at $w = 0.82$, reproducing the matched-horizon boundary almost exactly—a positive robustness check on the boundary itself. The *drift-sign* observable, however, was *inconclusive*: the restoring drift in the contested band is weakly positive across all w with no genuine zero-crossing, so it does not localise w_{crit}. The reason is basin geometry: PPO initialises above the cooperative optimum (S≈250 vs S≈150), and collapses often plunge through the contested band too quickly to be sampled, biasing the conditioned drift toward survivors. We therefore do *not* claim a magnitude-level prediction; the boundary magnitude remains calibrated, as stated in Table 4. The mechanism section stands on its sign-level and functional-form predictions, which are unaffected. The driver and analysis (with the decision rule encoded) are released with the code so the test can be re-run with alternative observables (e.g. a longer horizon matched to $w_{\text{crit}} = 0.666$, or a basin-flow statistic).

D Robustness: Exploration Noise and Heterogeneous Populations

Two further pre-registered experiments probe the boundary along axes the main analysis holds fixed: the exploration noise of the learner, and the homogeneity of the population. Both reuse the identical substrate, PPO configuration, and collapse label.

Exploration noise (entropy coefficient). The mean-field account (Section 6) predicts that w_{crit} is set by the signal-to-noise ratio of the policy gradient, so increasing the exploration term should raise it. We sweep the PPO entropy coefficient at the structural condition (α=β=1, λ=1, 300k, 20 seeds, $w \in \{0.5, \ldots, 0.9\}$). The boundary rises monotonically with exploration noise (Figure 7, Table 5): $w_{\text{crit}} = 0.685$ at ent=0 (reproducing the null baseline exactly), 0.690 at 0.001, 0.715 at 0.01, and beyond 0.05 the system collapses at *every* tested w (w_{crit} extrapolates above 1). This is direct empirical support for the signal-to-noise mechanism: the location of the boundary is not a function of the behavioral bias alone but also of the exploration noise that competes with the diluted cooperative signal, exactly as (9) predicts.

Table 5: Entropy-coefficient ablation (α=β=1, λ=1, 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs.

entropy coef.	w_{crit} (95% CI)
0.000	0.685 [0.656, 0.710]
0.001	0.690 [0.667, 0.715]
0.010	0.715 [0.689, 0.743]
0.050	> 1 (collapse at all w)

Heterogeneous cognitive profiles. We next ask whether a single loss-averse node destabilises an otherwise risk-neutral network, and whether its network position matters. We assign λ=7 to selected nodes and λ=1 to the rest (α=β=0.88, 300k, 20 seeds, $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$). The effect is large and strongly position-dependent (Figure 8, Table 6). A single *downstream* loss-averse node

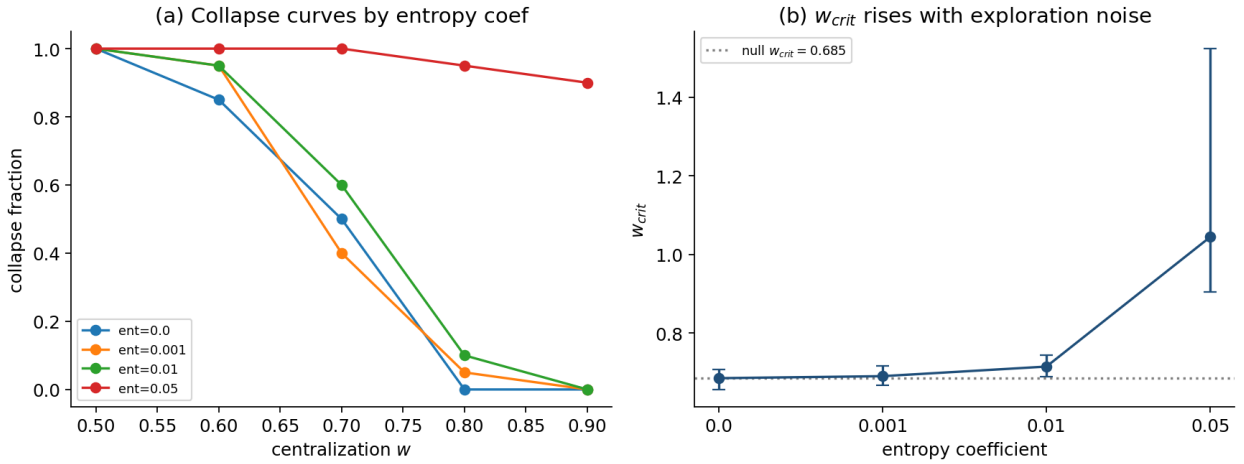


Figure 7: Exploration-noise ablation. (a) Collapse curves shift right as the entropy coefficient grows. (b) w_{crit} rises monotonically with exploration noise, corroborating the signal-to-noise mechanism (Section 6); ent=0 reproduces the null boundary 0.685.

drives $w_{\text{crit}} = 0.853$ $[0.849, 0.858]$—statistically indistinguishable from the *fully* loss-averse population ($w_{\text{crit}} = 0.851$ $[0.847, 0.856]$): one irrational node downstream poisons the entire network. A single *upstream* loss-averse node shifts the boundary far less, to $w_{\text{crit}} = 0.726$ $[0.689, 0.763]$. Loss aversion therefore cascades upstream—a downstream node's bias propagates through the ordering chain and dominates the system boundary—but does not cascade downstream, a directional asymmetry consistent with the bullwhip structure of the substrate. The practical implication is sharp: in a mixed human–agent supply chain, the system-level centralization requirement is governed by the most downstream loss-averse participant, not by the population average.

Table 6: Heterogeneous-population sweep (α=β=0.88; loss-averse nodes at λ=7, others λ=1; 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs. Homogeneous references: all risk-neutral ≈ 0.586, all loss-averse 0.851.

Loss-averse nodes	w_{crit} (95% CI)
one upstream	0.726 [0.689, 0.763]
one downstream	0.853 [0.849, 0.858]
two downstream	0.851 [0.847, 0.856]
all five (homogeneous)	0.851 [0.847, 0.856]

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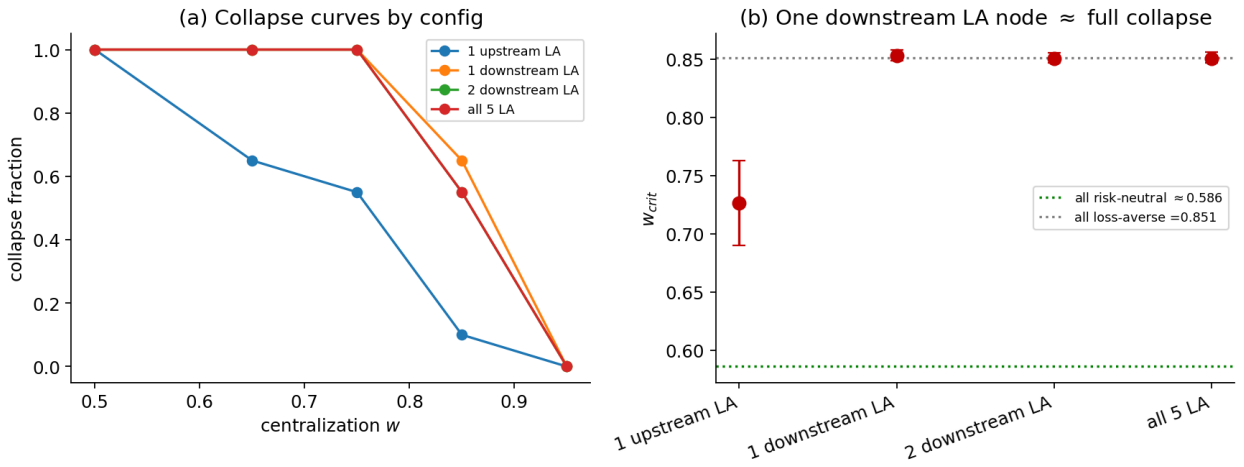


Figure 8: Heterogeneous cognitive profiles. (a) Collapse curves: a single downstream loss-averse node (orange) tracks the fully loss-averse population (red), while a single upstream node (blue) is far milder. (b) w_{crit} by configuration: one downstream loss-averse node already reaches the homogeneous loss-averse boundary; loss aversion cascades upstream but not downstream.

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